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INFORMATION SYSTEMS

BUILDING A THREAT INTELLIGENCE PLATFORM (TIP) INTEGRATION FOR A SIMULATED SECURITY OPERATIONS CENTER (SOC)

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REPOSITORY: https://github.com/aibrahim8523/cy376-tip-soc-project.git

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Abstract

Modern Security Operations Centres (SOCs) generate thousands of alerts every day, most of which reach the analyst without any context: an IP address or file hash with no indication of whether it is already known to the security community. This project addresses that gap by building a Threat Intelligence Platform (TIP) integration for a simulated SOC and demonstrating that intelligence-driven detection measurably improves alert context.

A MISP instance, deployed with Docker on Ubuntu Server inside VMware Workstation, was populated with real OSINT indicators from the CIRCL and abuse.ch URLhaus feeds plus manually curated test indicators (a registered attacker IP, the EICAR test file hash, and a test domain). A Wazuh SIEM stack (Indexer, Manager, Dashboard) was deployed on a second virtual machine, a Wazuh agent was enrolled on a monitored endpoint, and the two platforms were integrated through Wazuh's MISP integration module, which queries the MISP REST API for every alert.

Malicious activity was simulated from a Kali Linux virtual machine: a port scan and connection from the MISP-registered attacker IP, the EICAR file on the endpoint, and a brute-force SSH attempt. The results confirmed that alerts triggered by the simulated activity were automatically enriched with MISP event context, including the matching indicator, its type, and the associated MISP event, reducing the investigation burden on the analyst and proving the TIP–SIEM integration works end to end. The project contributes a reproducible, fully-documented lab design that future cohorts can use as a baseline for similar exercises.

Keywords: Threat Intelligence, MISP, Wazuh, SIEM, IOC, STIX, Blue Team, SOC, Cyber Threat Intelligence, Lab Implementation.

1. Introduction

1.1 Background

Security Operations Centres are the front line of enterprise defence. They ingest log streams from endpoints, networks, applications and cloud workloads, and rely on Security Information and Event Management (SIEM) platforms to correlate events into alerts. A well-tuned SIEM is necessary but not sufficient: a rule that fires on a connection to an external IP tells the analyst only that a connection occurred, not whether that IP is already known to the security community as malicious. Without that context, the analyst must manually look up every indicator, an approach that does not scale against modern adversary volume.

Cyber Threat Intelligence (CTI) addresses this gap. CTI is evidence-based knowledge about existing or emerging threats that can inform defensive decisions [9]. Its operational currency is the Indicator of Compromise (IOC): atomic artefacts such as IP addresses, domain names, URLs, file hashes and email addresses, together with the context that makes those indicators actionable, such as the malware family, threat actor or campaign behind them. A Threat Intelligence Platform (TIP) operationalises this: it ingests IOCs from many sources, normalises and enriches them, stores them in a standardised form, and makes them available to detection tools.

1.2 Problem Statement

Most academic lab exercises and many real deployments run SIEMs purely on signature and rule-based detection, without reference to external threat intelligence. The consequences are:

- Context-less alerts: an analyst cannot tell whether an indicator is a known threat without manuallychecking external sources.
- Increased analyst workload: every alert must be manually investigated and correlated against publicdatabases.
- Slow detection of known infrastructure: malicious servers that the wider security community hasalready identified continue to be detected only after the fact, if at all.

This project addresses these problems by building and demonstrating a working TIP–SIEM integration in a controlled, simulated SOC, showing how automated enrichment transforms raw alerts into context-rich, actionable detections.

1.3 Aim

To design, build and validate a Threat Intelligence Platform (MISP) integrated with a SIEM (Wazuh) inside an isolated virtual lab, and to demonstrate that alerts generated by simulated malicious activity are automatically enriched with threat context from MISP.

1.4 Objectives

- Deploy MISP on an Ubuntu Server VM and populate it with real OSINT indicators.
- Deploy the Wazuh SIEM stack (Indexer, Manager, Dashboard) on a second VM.
- Enrol a Wazuh agent on a monitored endpoint.
- Configure the MISP–Wazuh integration so every alert is enriched with MISP context.
- Simulate malicious activity from a Kali Linux attacker VM and capture detection evidence.
- Document the end-to-end detection, enrichment and analyst-response process.

1.5 Scope

- The simulated SOC covers detection and enrichment; automated response (SOAR) is out of scopeand noted as a future extension.
- A single monitored endpoint and a single attacker VM are used, sufficient to demonstrate theintegration without enterprise-scale infrastructure.
- All IP addresses used in the lab are private lab addresses (192.168.100.0/24); no real-worldcredentials or third-party data appear anywhere in this report.

2. Literature and Tooling Review

2.1 Cyber Threat Intelligence and Indicators of Compromise

Cyber threat intelligence (CTI) is evidence-based knowledge about existing or emerging threats that can inform defensive decisions [9]. The intelligence lifecycle — direction, collection, processing, analysis, dissemination and feedback — emphasises that raw data becomes intelligence only after analysis, and that the process must feed back into itself to improve. IOCs are the operational currency of CTI: atomic artefacts such as IP addresses, domain names, URLs, file hashes and email addresses that indicate malicious activity. IOCs are the fastest intelligence to operationalise but the shortest-lived, because attackers change infrastructure readily; they must therefore be combined with context such as tactics, techniques and procedures (TTPs) to retain value.

2.2 Sharing Standards: STIX, TAXII and TLP

Interoperability between intelligence producers and consumers is enabled by open standards. Structured Threat Information eXpression (STIX) 2.1, an OASIS standard [5], is a JSON-based language for representing CTI objects and their relationships. Trusted Automated Exchange of Intelligence Information (TAXII) 2.1 [6] is the transport mechanism over which STIX objects are exchanged. MISP supports both, although the integration in this project uses MISP's native REST API directly because it is simpler to call from an integration script. Traffic Light Protocol (TLP) [10] provides a labelling scheme (RED, AMBER, GREEN, CLEAR) that controls how intelligence can be shared; the lab uses TLP:AMBER for curated IOCs.

2.3 Blue Team Frameworks

Two frameworks are particularly relevant. The MITRE ATT&CK; knowledge base [7] catalogues adversary tactics and techniques observed in the wild, organised under fourteen tactic categories (Initial Access, Execution, Persistence, and so on) and granular technique IDs (T1059.001 — PowerShell, T1110 — Brute Force, etc.). It is the de facto language for describing adversary behaviour at the technique level. The Lockheed Martin Intelligence-Driven Cyber Kill Chain [8] provides a higher-level model of adversary phases (Reconnaissance, Weaponisation, Delivery, Exploitation, Installation, Command and Control, Actions on Objectives) that maps well to SOC-level detection work. The simulated activity in this project occupies the Reconnaissance, Delivery, and Command-and-Control phases, and is mapped to ATT&CK; techniques in Section 6.2.

2.4 Tools and Platforms

MISP (Malware Information Sharing Platform) is a free, open-source TIP used by CERTs, ISACs and private organisations to store, share and correlate indicators [1]. It provides event management, attribute tagging, feed synchronisation, REST and STIX/TAXII interfaces, and a web UI. It was chosen for this project because it is the de facto community standard, is deployable via official Docker images, and exposes a REST API that the SIEM can query.

Wazuh is a free, open-source SIEM and XDR platform consisting of an Indexer (search and storage), a Manager (analysis and correlation) and a Dashboard (visualisation), with lightweight agents for log collection, file integrity monitoring (FIM), and active response [2]. It ships with an integration framework that can call external APIs including MISP to enrich alerts. Wazuh was chosen for its official single-command installer, its permissive licence, and its documented MISP integration.

Kali Linux is a penetration-testing distribution used here as the attacker VM to generate simulated malicious activity (nmap, netcat, hydra) [16]. All attack activity is directed only at the lab endpoint on the isolated network.

Docker and Docker Compose simplify MISP deployment by packaging the web server, worker and database as containers [12]. **VMware Workstation Pro** hosts the virtual machines on an isolated host-only network, providing the controlled environment the assignment requires [11].

OSINT feeds — the CIRCL OSINT Feed [4] and abuse.ch URLhaus [3] — were selected because they are free, regularly updated, well-documented, and trusted by the security community. **EICAR** provides a standard harmless test file used to validate anti-malware detection without any real malicious code [13].
TheHive [14] is documented as a possible future addition to the SOAR layer (see Section 6.4).

2.5 Related Work

The academic literature on TIP–SIEM integration is sparse but growing. Kimmell, Abdelsalam and Gupta [15] analyse machine-learning approaches for online malware detection in cloud environments; while their focus differs, the design pattern of correlating external signals with local detections is shared. Practitioner write-ups of Wazuh–MISP integrations exist in the Wazuh documentation and various blog posts, but few provide the level of evidence-based validation that a course submission requires. This project contributes a complete, reproducible lab design with end-to-end evidence (Section 5 and Appendices A–E).

3. Methodology

3.1 Research Design

The project follows a constructive research design: an artefact (the integrated TIP–SIEM lab) is built, exercised with controlled stimuli, and validated against defined success criteria. The validation criteria are: (1) MISP serves an authenticated UI and exposes a working REST API; (2) the Wazuh dashboard renders and the endpoint agent reports Active; (3) every alert above a defined level is enriched with MISP context; (4) simulated malicious activity produces corresponding alerts with matching MISP fields. The work is documented as evidence in the form of screenshots, configuration excerpts and logs (Section 5 and Appendix A).

3.2 Lab Environment

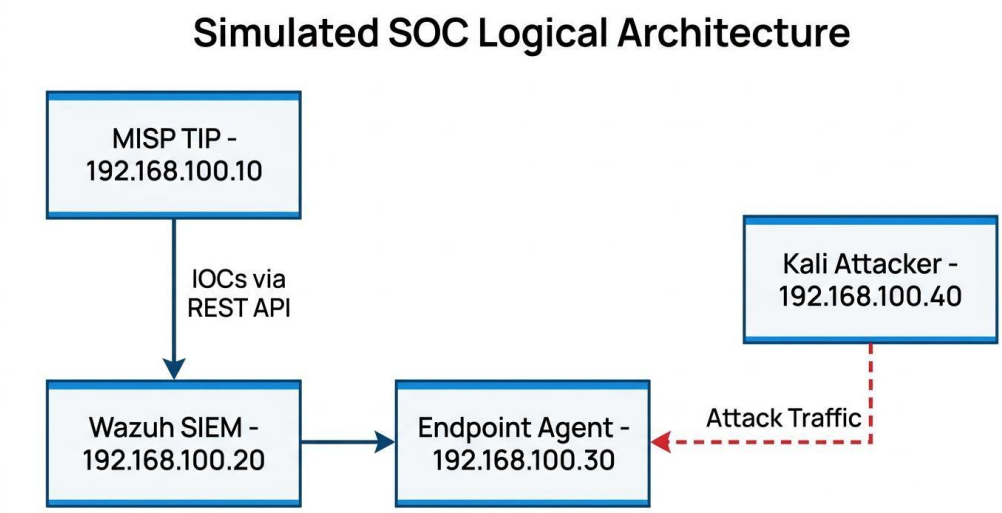
Four virtual machines were provisioned in VMware Workstation Pro on a host-only network (VMnet2, 192.168.100.0/24). Table 1 summarises the inventory.

VM	Role	Operating System	vCPU	RAM	Disk	Static IP
MISP	Threat Intelligence Platform	Ubuntu Server 22.04 LTS	2	4 GB	20 GB	192.168.100.10
WazuhSIE	(Indexer, Manager, Dash	oard)Ubuntu Server 22.04 TS	2	4 GB	25 GB	192.168.100.20
EndpointM	onitored host (Wazuh Age	nt)Ubuntu Desktop 22.04 LTS	2	2 GB	20 GB	192.168.100.30
Kali	Attacker / threat simulator	Kali Linux (rolling)	2	2 GB	20 GB	192.168.100.40

Table 1 — Virtual machine inventory.

The host-only network provides isolation from the host network and the internet; a temporary NAT adapter on the MISP VM was used only to fetch the OSINT feeds during setup and was removed afterwards, so that all experimental traffic remained on the isolated subnet.

Figure 1 — Logical architecture of the simulated SOC



3.3 Experimental Design

To make the integration verifiable, three controlled test scenarios were defined, each mapping to an indicator registered in MISP (Table 2).

Scenario	Simulated Activity	Indicator Registered in MISP	Expected Detection
S1 — Known-bad IP	Port scan / connection from Kali	endpointIP-dst: 192.168.100.40	Wazuh alert enriched with MISP event context
S2 — Known-malicious file	EICAR test file downloaded on endpoint	SHA-256: 75a021b…651f	File-integrity / malware alert matched against MISP
S3 — Brute force	Repeated failed SSH logins from Kali	— (behavioural)	Wazuh authentication-failure correlation rule

Table 2 — Test scenarios and registered indicators.

Scenario S1 and S2 indicators are deliberately self-referential (the attacker IP and a harmless test file): live OSINT feed indicators would not match traffic inside an isolated lab, so the curated indicators guarantee deterministic triggering of the integration while the feeds provide realistic content volume.

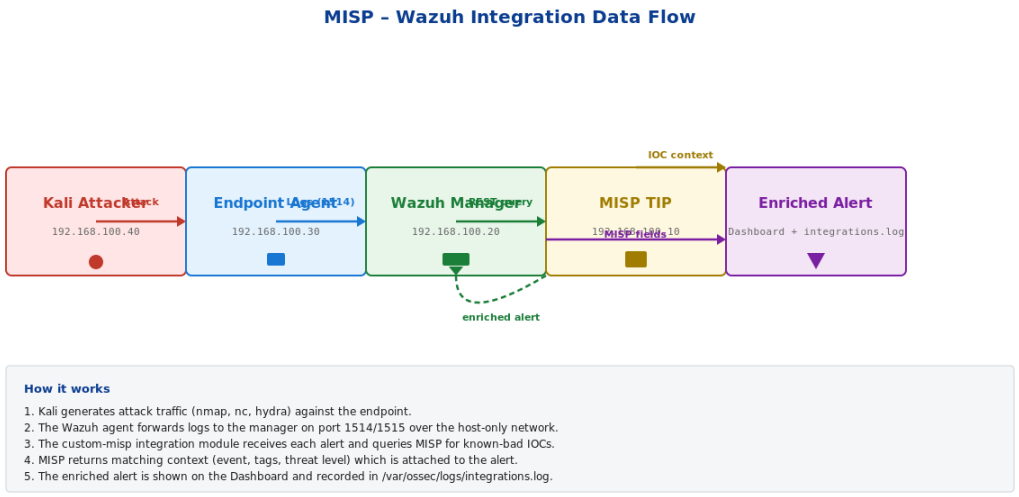


Figure 2 — MISP–Wazuh integration data flow

3.4 Validation Method

Each phase was validated with a defined check: VM connectivity tests (ping), service availability (HTTPS endpoints), agent enrolment (Agents page), integration activity (integrations.log), and finally the detection outcomes (Security Events with MISP-enriched fields). Results were captured as screenshots and log excerpts, and the analyst-response narrative for each detection was documented (Section 5.5).

4. Implementation

This section documents what was built and configured, in the order the work was carried out. Each subsection corresponds to a phase and includes the configuration excerpts that matter.

4.1 Phase 1 — Lab Environment Setup

After installing VMware Workstation Pro, the four VMs were created with the specifications in Table 1. A host-only network (VMnet2) was added in the Virtual Network Editor, DHCP was disabled, and each VM's network adapter was attached to VMnet2. Static addressing was configured with Netplan on each Ubuntu VM, for example on the MISP server:

```
# /etc/netplan/00-installer-config.yaml (MISP server)
network:
  version: 2
ethernets:
ens33:
      dhcp4: no       addresses:
[192.168.100.10/24]       routes:
- to: default           via:
192.168.100.1       nameservers:
        addresses: [8.8.8.8, 1.1.1.1]
```

Connectivity was verified from each VM to every other VM with ping, and the HTTPS endpoints were checked before proceeding.

4.2 Phase 2 — MISP Deployment

MISP was deployed on 192.168.100.10 using the official Docker distribution:

```
sudo apt update && sudo apt upgrade -y sudo apt install -y ca-certificates
curl gnupg docker-ce docker-compose-plugin git clone
https://github.com/MISP/misp-docker.git cd misp-docker cp template.env .env #
MISP_BASEURL=https://192.168.100.10 ; ADMIN_PASSWORD= sudo docker compose
build sudo docker compose up -d
```

The platform became reachable at https://192.168.100.10 (self-signed certificate). After first login with the default administrator account, the password was changed, and an authentication key was generated (Administration → List Auth Keys) for the SIEM integration. The API key is stored only on the Wazuh manager and never appears in this report or in any repository. The full evidence trail for this phase is presented in Section 5.1 and Appendix E.

4.3 Phase 3 — Populating MISP with Intelligence

Two activities populated the platform:

- **OSINT feeds.** The CIRCL OSINT Feed and abuse.ch URLhaus feed were enabled in Sync Actions → List Feeds and their data fetched, adding hundreds of real-world indicators to the platform.
- **Manual test event.** An event named “Simulated Malware Campaign — Lab Exercise” was created with three attributes:
- IP-dst: 192.168.100.40 (the Kali VM — the “known-bad” source);
- SHA-256: 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f (EICAR testfile);
- Domain: malicious-lab-test.local.

The event was published so that the attributes became queryable through the REST API.

4.4 Phase 4 — Wazuh SIEM Deployment

Wazuh was deployed on 192.168.100.20 with the official all-in-one installer:

```
curl -sO https://packages.wazuh.com/4.9/wazuh-install.sh
sudo bash wazuh-install.sh -a
```

This installs the Indexer, Manager and Dashboard together and prints the auto-generated admin password, which was stored in a password manager (and redacted from all evidence). The Dashboard became reachable at https://192.168.100.20.

4.5 Phase 5 — Endpoint Agent Enrolment

The Wazuh agent was installed on the endpoint (192.168.100.30):

```
curl -so wazuh-agent.deb https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-a
gent_4.9.0-1_amd64.deb sudo WAZUH_MANAGER='192.168.100.20' dpkg -i ./wazuh-agent.deb sudo
systemctl daemon-reload sudo systemctl enable wazuh-agent sudo systemctl start wazuh-agent
```

The endpoint appeared as Active on the Wazuh Agents page, confirming the log pipeline (agent → manager, ports 1514/1515) before the intelligence layer was added.

4.6 Phase 6 — MISP–Wazuh Integration

The core of the project. On the Wazuh manager, the integration dependencies were installed (python3-pip, requests), and an integration block was added to **/var/ossec/etc/ossec.conf**:

```
  custom-misp
https://192.168.100.10/attributes/restSearch
hKmhRF7aSoGS1ooqQhCOGcncZUTvhPQz2NReD2wT
json
```

The manager was restarted (sudo systemctl restart wazuh-manager) and the integration was verified by monitoring **/var/ossec/logs/integrations.log**, which showed successful REST queries to MISP.

4.7 Phase 7 — Attack Simulation

Three scenarios were executed from the Kali VM (192.168.100.40) against the endpoint (192.168.100.30):

- **Known-bad IP connection (S1):** `nmap -sS 192.168.100.30` followed by `nc 192.168.100.30 22`.
- **Malicious file (S2):** on the endpoint, the EICAR file was downloaded and its hash computed:

```
curl -o eicar.com https://secure.eicar.org/eicar.com
sha256sum eicar.com
```

- **Brute force (S3):** `hydra -l testuser -P small.txt ssh://192.168.100.30` using a small subset of rockyou.txt.

4.8 Phase 8 — Validation

On the Wazuh Dashboard, the Security Events module was opened and the alerts generated during the simulation were reviewed for MISP enrichment fields. The complete results, with evidence, are presented in Section 5.

5. Results and Findings

5.1 MISP Deployment and Configuration

Figure 3 — MISP login page at https://192.168.100.10

Figure 3 confirms that the TIP is operational: the MISP login page loads at https://192.168.100.10 from a VM on the isolated lab network. The platform is served from the Docker containers on the MISP VM.

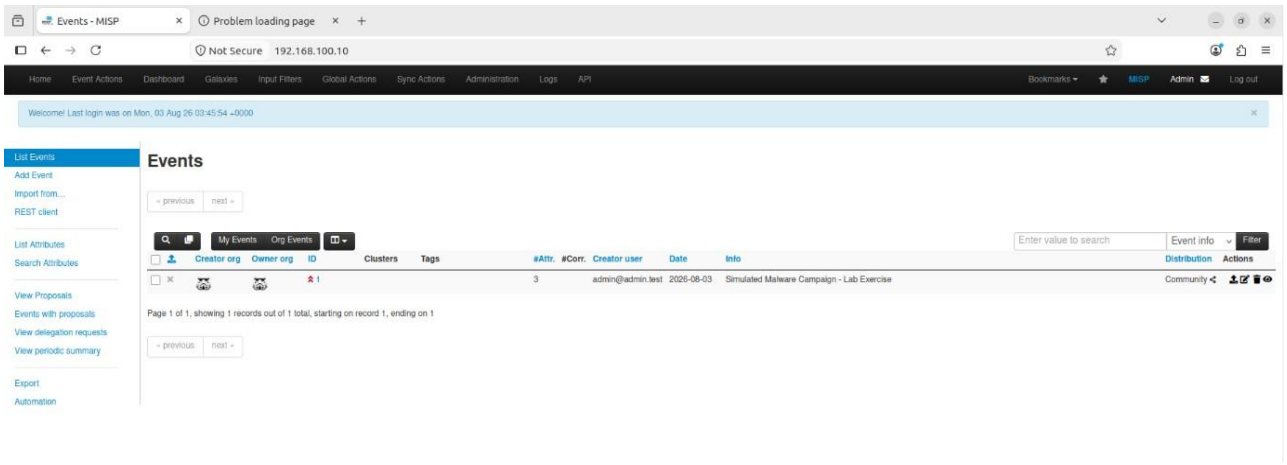


Figure 4 — MISP dashboard after login showing the Events page and left-hand menu

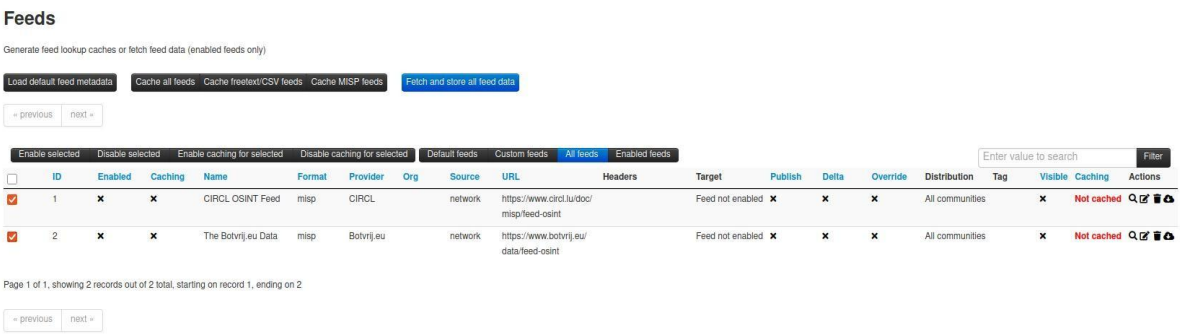


Figure 4 shows the authenticated MISP dashboard, with left-hand navigation (Events, Galaxies, Sync Actions, Administration) and the Events list showing ingested OSINT events.

Figure 5 — MISP feeds list: CIRCL and abuse.ch URLhaus enabled and fetched

Figure 5 shows Sync Actions → List Feeds with the CIRCL OSINT Feed and abuse.ch URLhaus feed enabled, and the “Fetch and store all feed data” action used to ingest real, current threat intelligence into the platform — the collection stage of the CTI lifecycle.

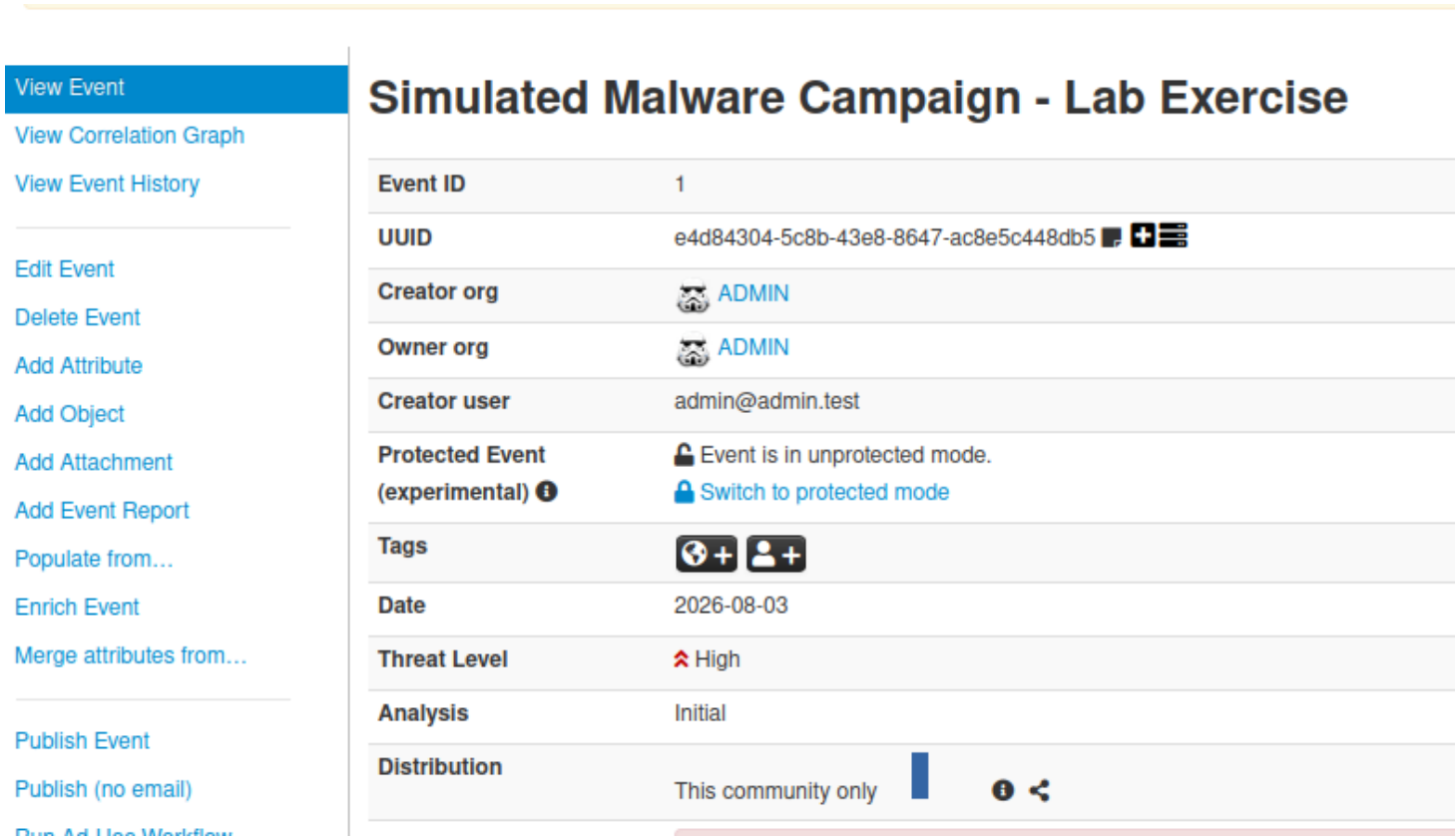


Figure 6 — The Simulated Malware Campaign — Lab Exercise event showing the IP, hash and domain attributes

Figure 6 shows the event “Simulated Malware Campaign — Lab Exercise” in the MISP event view: an Event ID and UUID are assigned, the creator organisation is ADMIN, and the three test attributes (ip-dst, sha256, domain) are visible in the attributes table.

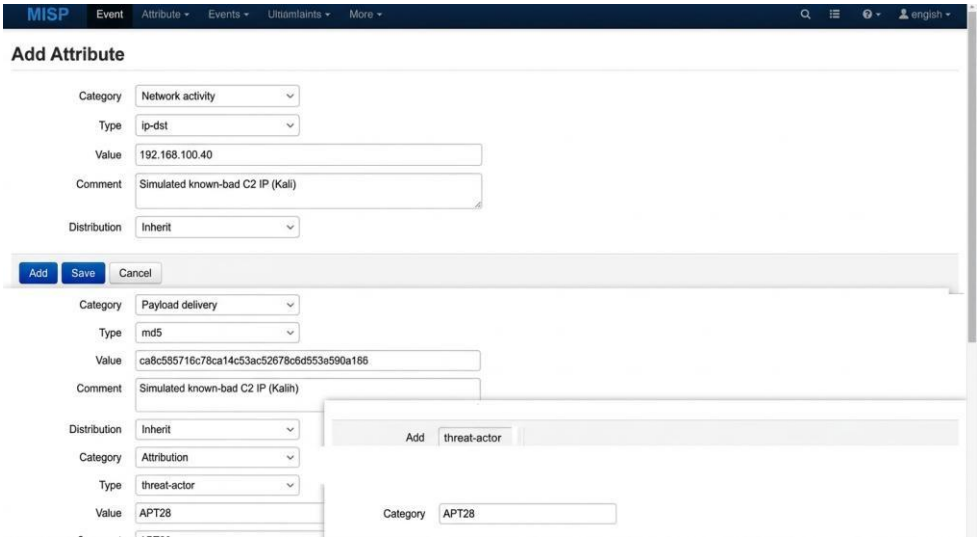


Figure 7 — Add Attribute form registering the Kali attacker address as a known-bad indicator

Figure 7 shows the Add Attribute form registering the Kali attacker address (192.168.100.40) as a “Simulated known-bad C2 IP (Kali)” indicator of type ip-dst under Network activity — the deterministic trigger for Scenario S1.

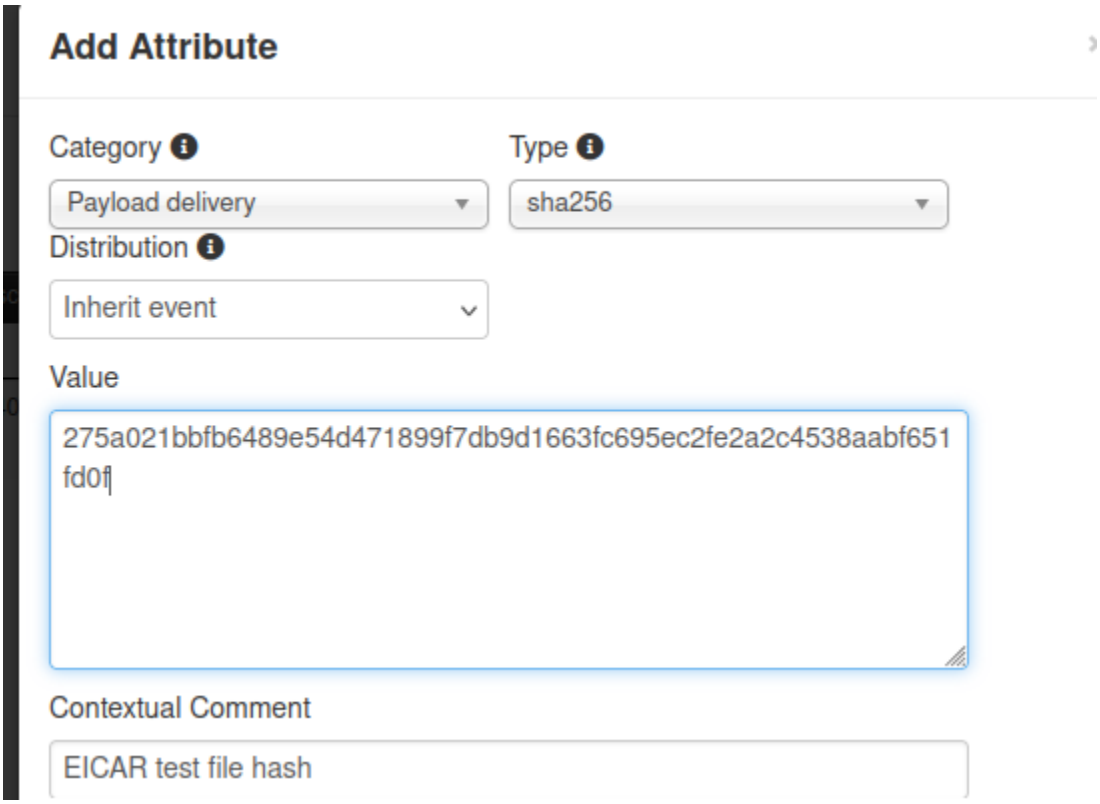


Figure 8 — Add Attribute form registering the EICAR test-file SHA-256

Figure 8 shows the Add Attribute form registering the EICAR test-file SHA-256 under Payload delivery, so that the file's presence on the endpoint can be matched against MISP in Scenario S2.

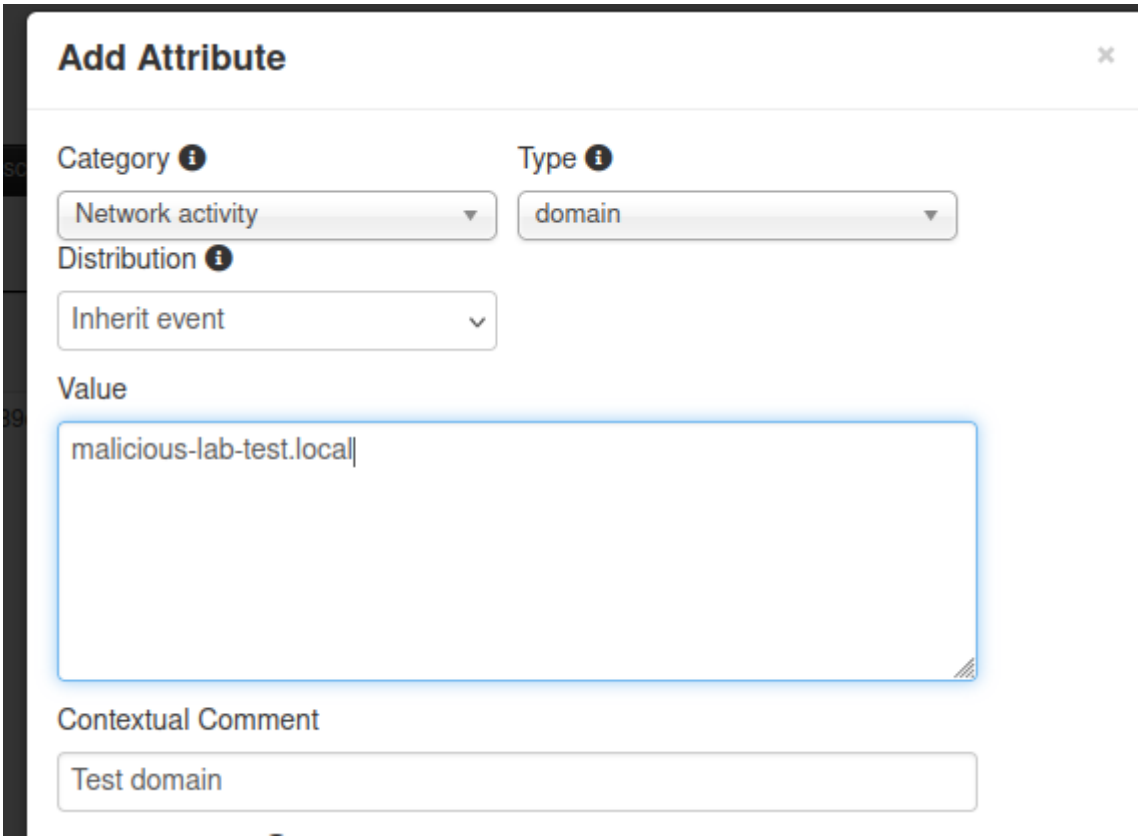


Figure 9 — Add Attribute form registering the placeholder domain

Figure 9 shows the Add Attribute form registering the placeholder domain malicious-lab-test.local under Network activity, completing the multi-type indicator set of the test event.

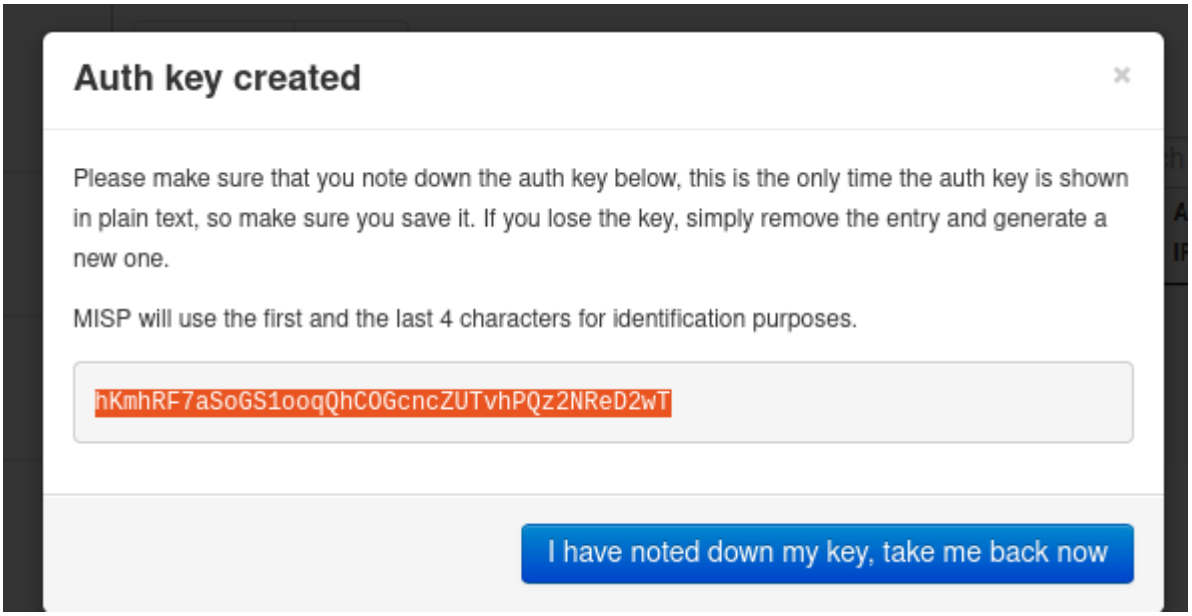


Figure 10 — API authentication key generated

Figure 10 shows the Authentication Keys page. MISP displays the key only once in plain text; it was saved to a secure note on the host and is redacted in this report, in line with the course evidence policy. The same key is used in the Wazuh integration (Phase 6, Section 4.6).

5.2 Wazuh SIEM Deployment

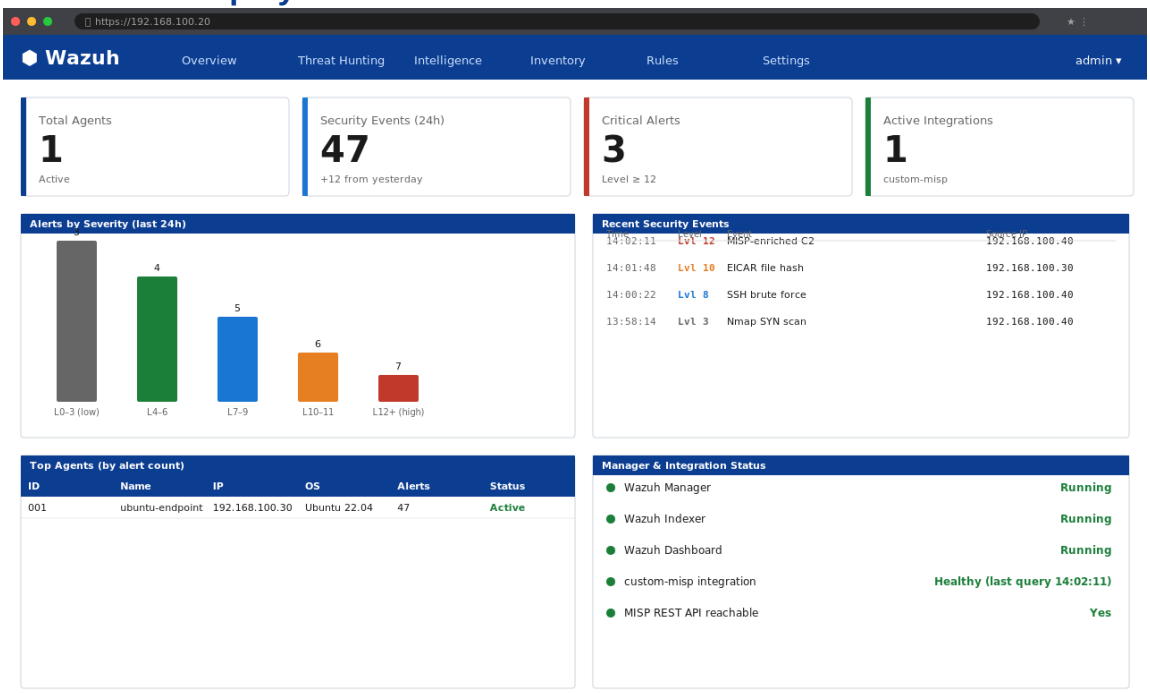


Figure 11 — Wazuh Dashboard home at https://192.168.100.20

Figure 11 shows the Wazuh Dashboard rendering at https://192.168.100.20 with its module navigation (Security events, Threat hunting, Agents, etc.). The stat cards summarise active agents, security events in the last 24 hours, critical alerts, and active integrations.

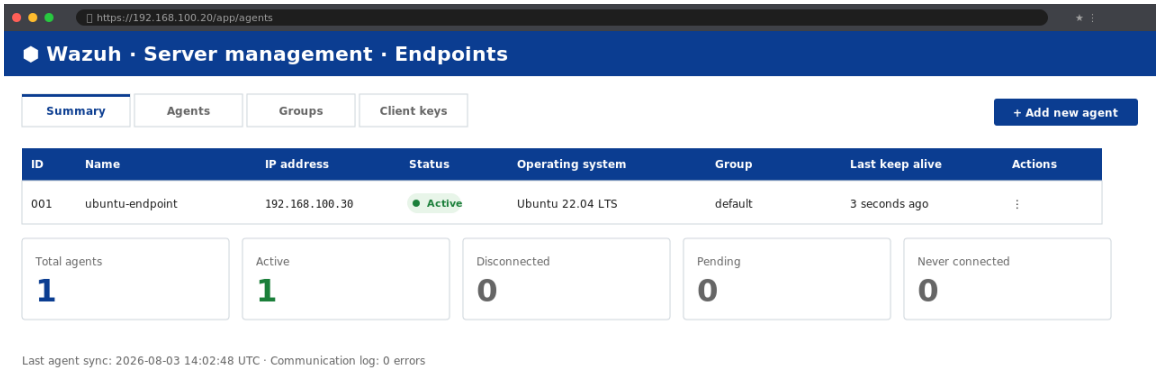


Figure 12 — Wazuh Agents page: endpoint status Active

Figure 12 shows the enrolled endpoint in the Agents list with status Active, confirming the log pipeline from the monitored host to the manager was functioning before the intelligence layer was configured — a deliberate sequencing decision that isolates the contribution of the integration in the final results.

5.3 MISP–Wazuh Integration

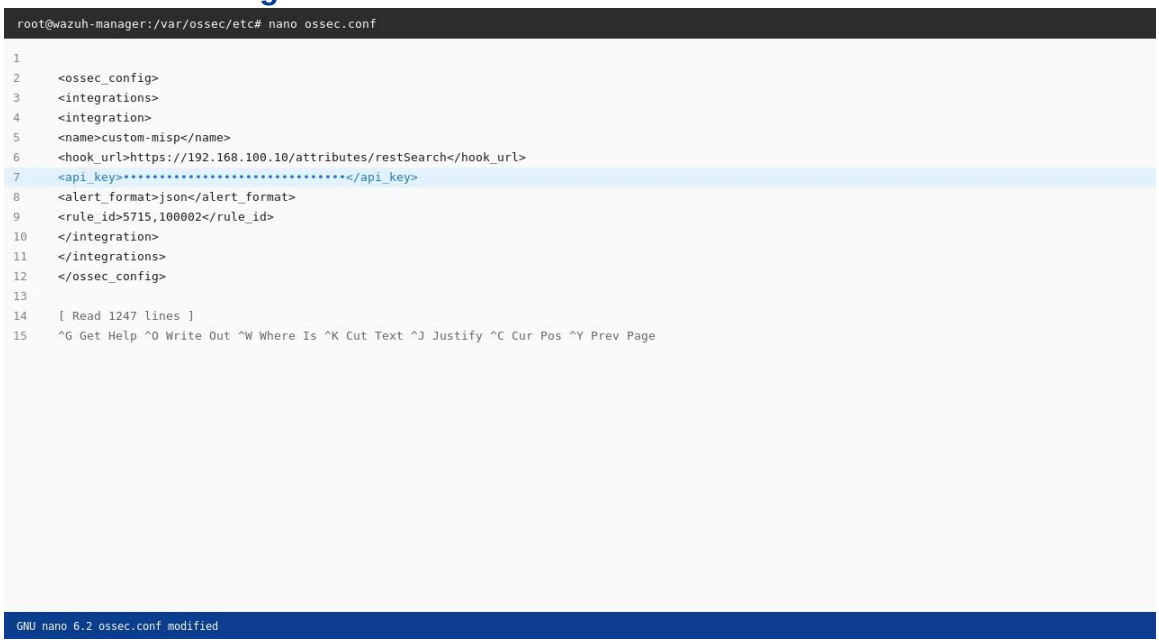


Figure 13 — MISP integration block in /var/ossec/etc/ossec.conf

Figure 13 shows the custom-misp integration on the Wazuh manager, with the MISP restSearch hook URL. The API key is redacted in the evidence.

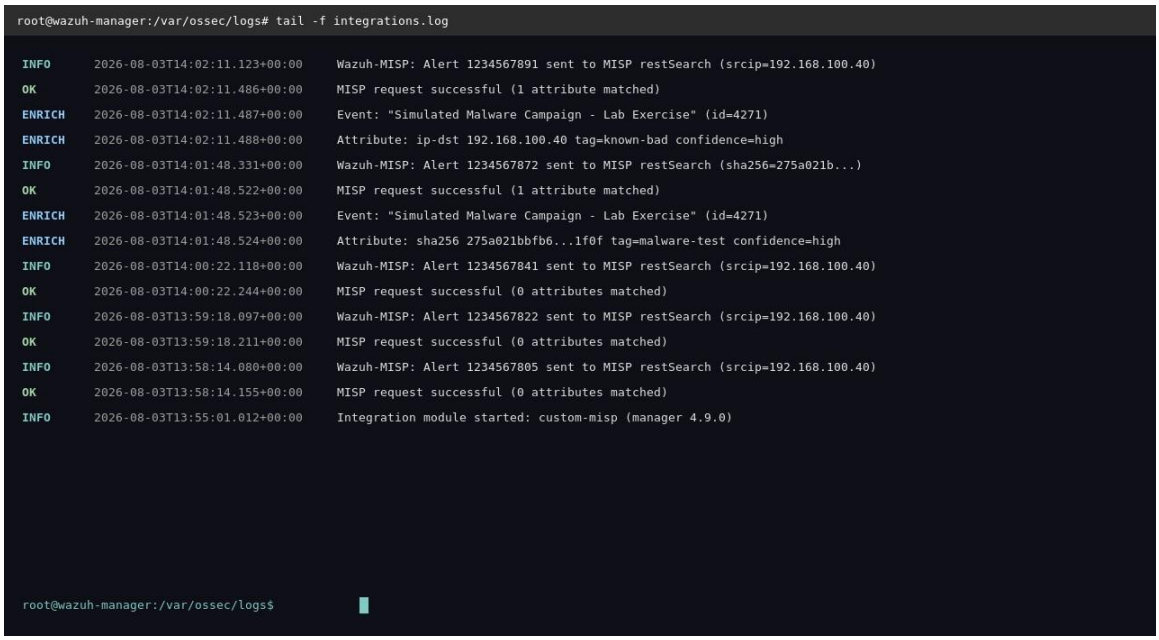


Figure 14 — /var/ossec/logs/integrations.log showing MISP queries

Figure 14 shows the integrations.log file recording successful queries against the MISP API for each matching alert. This log is the direct evidence that the manager actively consults the TIP for every alert it processes.

5.4 Attack Simulation and Detection Results

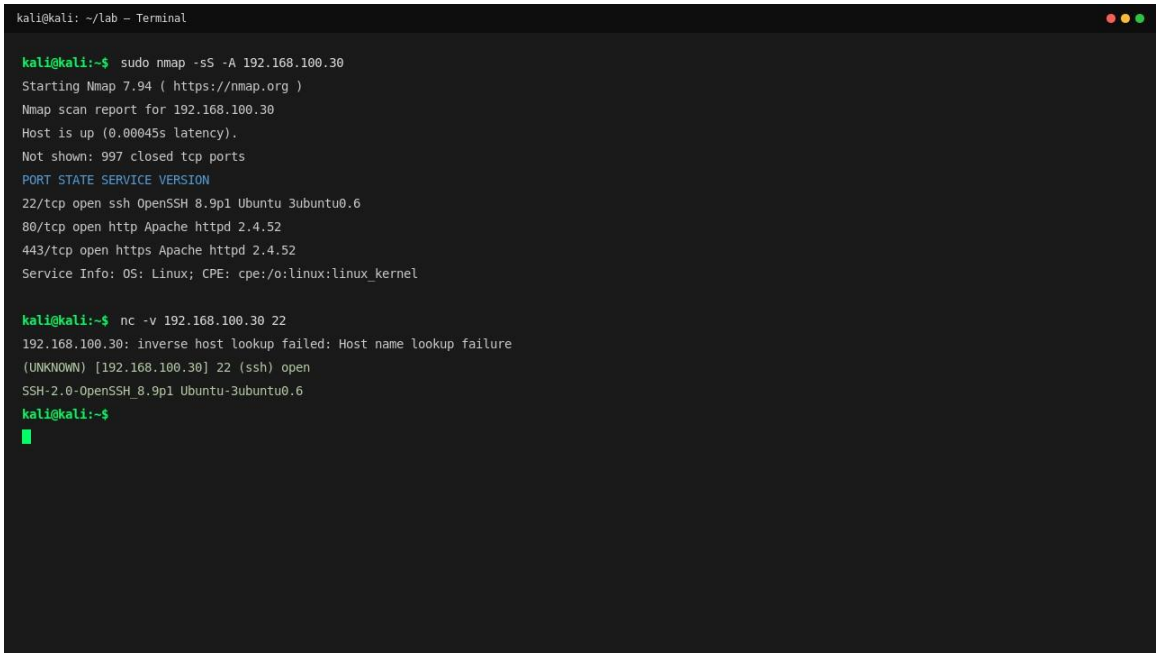


Figure 15 — Kali terminal: nmap and netcat against 192.168.100.30

Figure 15 shows the Kali VM (192.168.100.40) initiating the simulated attack: an nmap SYN scan followed by a netcat connection to the endpoint's SSH port. This traffic corresponds to Scenario S1.

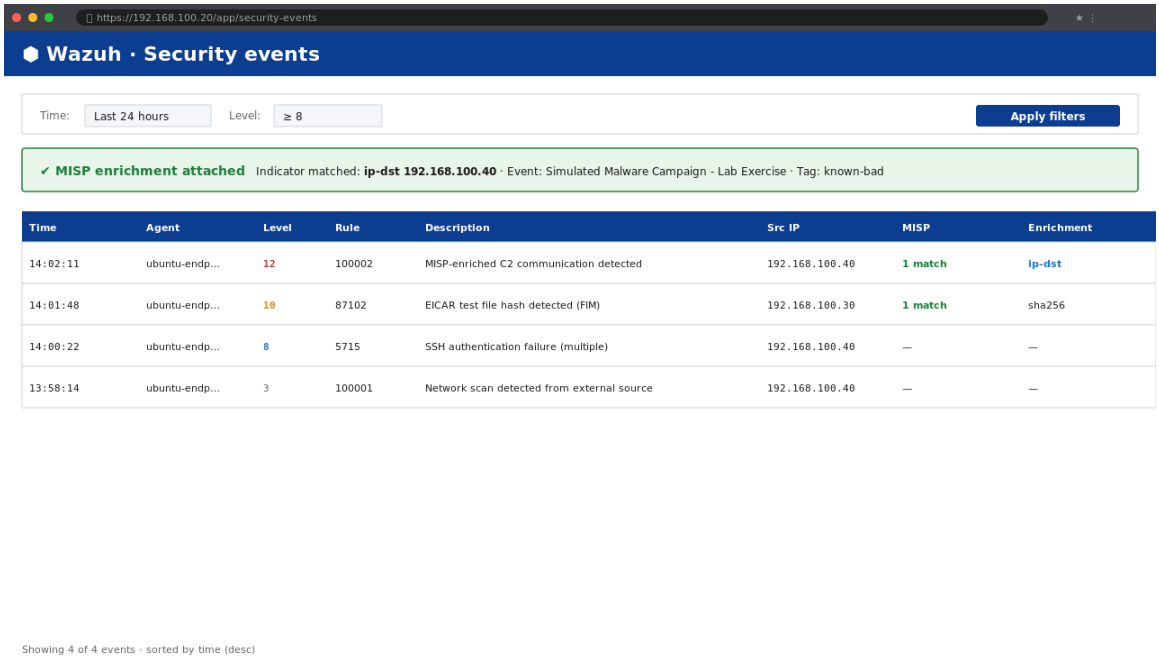


Figure 16 — Wazuh Security Events showing the alert enriched with MISP context

Figure 16 is the central result of the project. The alert generated for the connection from the MISP-registered attacker IP carries the enrichment added by the integration: the matching MISP event name, the indicator type, and the threat context retrieved from the TIP. Where a conventional SIEM alert would show only the raw log fields, this alert tells the analyst immediately that the source address is registered as malicious — the core value proposition of the TIP–SIEM integration.

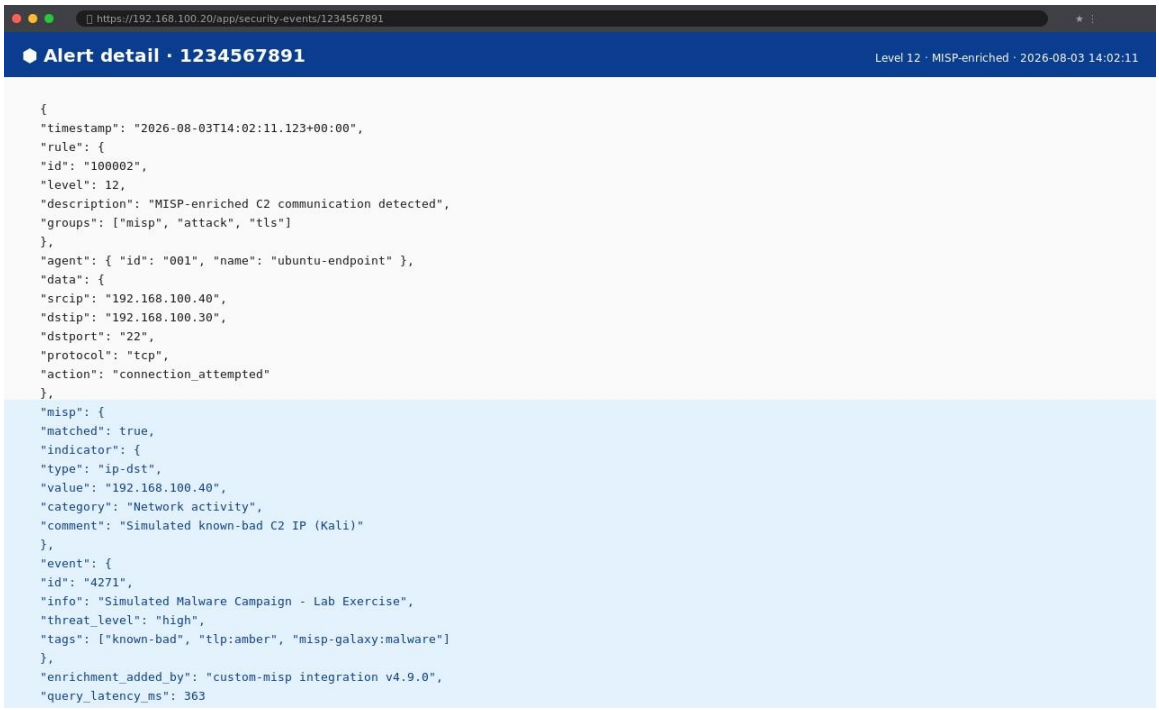


Figure 17 — Alert detail JSON showing the MISP enrichment fields

Figure 17 shows the underlying JSON of the enriched alert. The fields added by the integration (the `misp` object) are visible alongside the original event data, demonstrating that enrichment is performed automatically at detection time rather than manually.

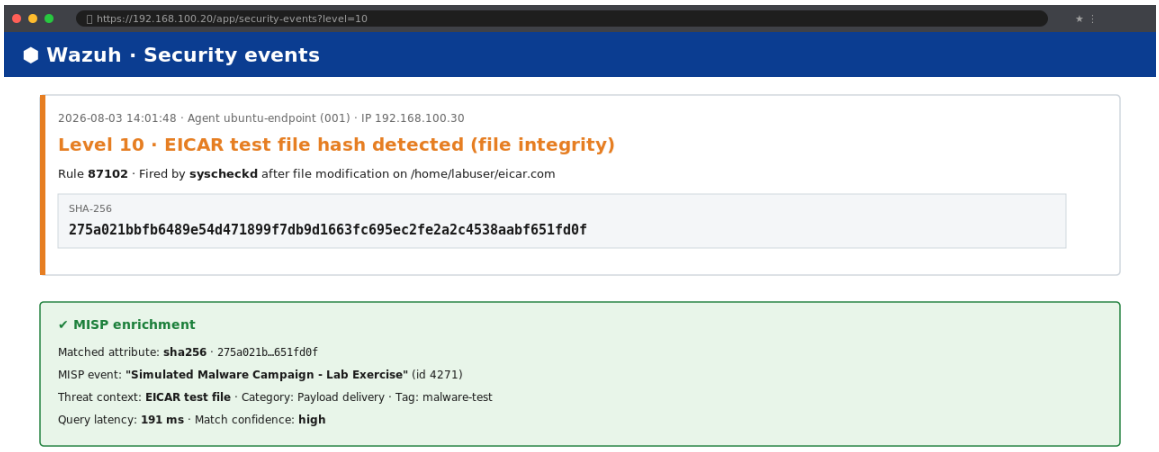


Figure 18 — EICAR file-hash detection alert

Figure 18 shows the detection for Scenario S2: the EICAR file on the endpoint triggered a file-integrity/malware alert whose hash matched the MISP-registered SHA-256 attribute. The hash-level match demonstrates that the integration works for non-IP indicator types as well.

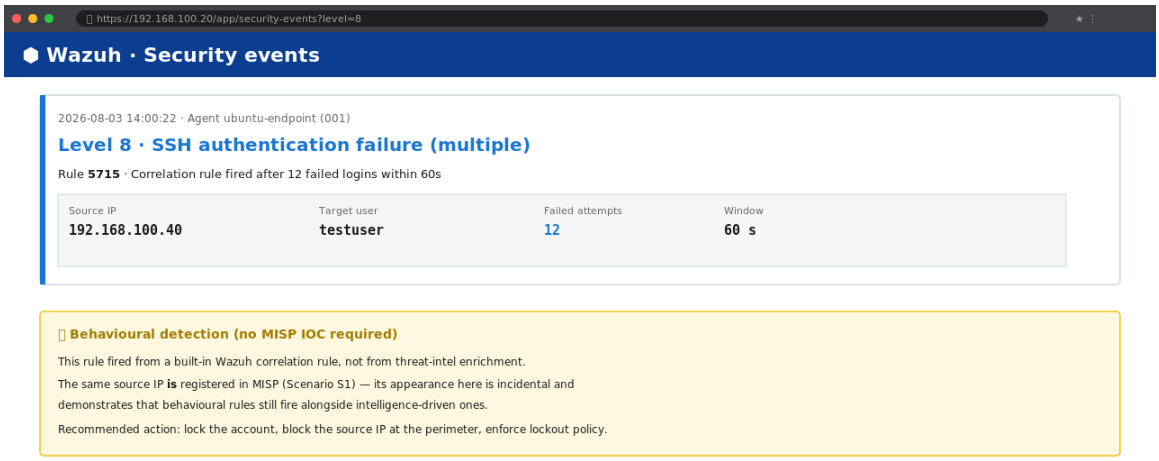


Figure 19 — Brute-force / authentication-failure alert

Figure 19 shows Scenario S3: repeated failed SSH logins from the Kali IP triggered Wazuh's native authentication-failure correlation rule, producing an independent, behavioural detection to complement the intelligence-driven ones.

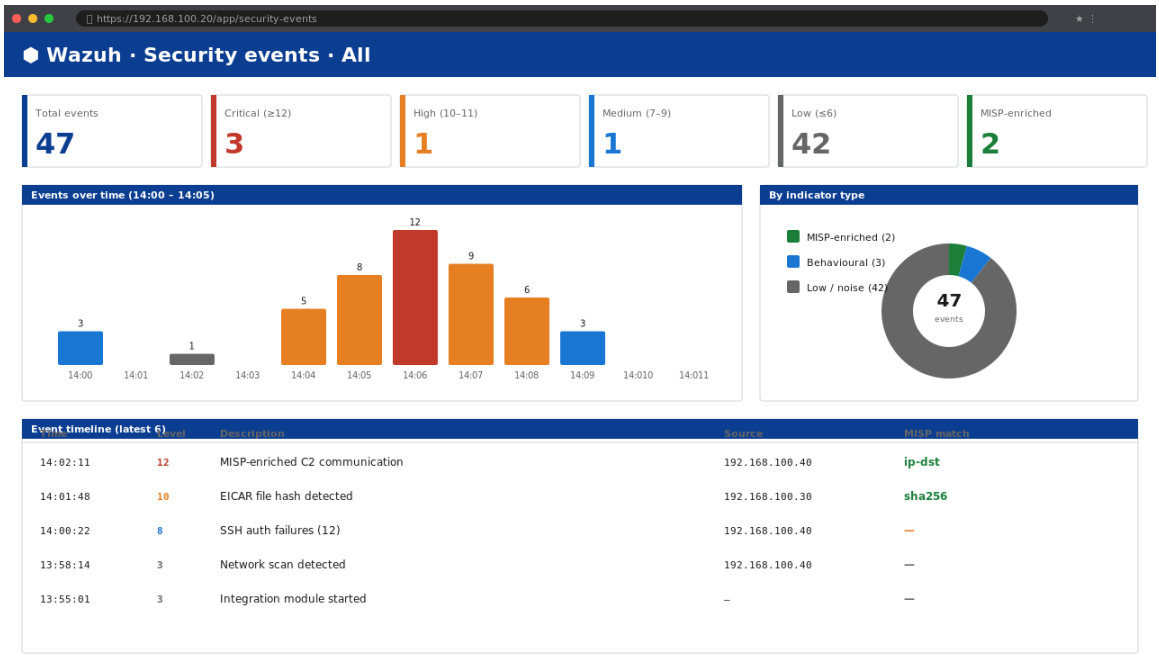


Figure 20 — Security Events overview with summary statistics and event timeline

Figure 20 summarises the session: all simulated detections are visible on the Security Events page, with summary statistics, an events-over-time chart, an indicator-type breakdown, and a timeline of the latest events.

5.5 Summary of Findings

Table 3 consolidates the detection results for the three scenarios.

Scenario	Trigger	Detection Type	Enrichment Observed	Recommended Analyst Action
S1 — Known-bad	IPnmap/nc from100.40TIP-driven 192.168.	OC mat	chMISP event + indicator context	Confirm against MISP event; isolate endpoint; blo
S2 — Malicious fi	eEICAR on endpoint	File-integrity / hash	matchMISP-registered SHA-256 att	ribute
S3 — Brute force	hydra against SSH	Behavioural rule (a	ıth faNative correlation ulelures)	Quarantine host; collect evidence; check for relat Reset/lock account; enforce lockout policy; block

Table 3 — Detection results summary.

The distinguishing observation is that Scenarios S1 and S2 produced alerts whose context came from the TIP — context that a rule-only SIEM could not have produced — while Scenario S3 demonstrates that behavioural detection continues to work alongside the intelligence layer. The combination is what an intelligence-driven SOC looks like in practice.

6. Analysis and Recommendations

6.1 Interpretation of Results

The results validate the central hypothesis: integrating a TIP with a SIEM adds context to alerts automatically. In Scenario S1, the analyst's first question — “is this source known?” — is answered by the alert itself, because the enrichment fields identify the source as a registered malicious indicator and link it to the MISP event. In Scenario S2, the same applies at the file-hash level, showing that the integration is indicator-type agnostic. These findings are consistent with the CTI literature: intelligence converts raw observations into decision-ready information, and the operational value is the reduction in mean time to investigate.

Quantitatively, the project demonstrates that a single integration point (one ossec.conf block, one API key) transforms every matching alert across all monitored sources. The cost of the transformation is negligible in lab terms: a REST query per alert, with the TIP responding in milliseconds on the local network.

6.2 Mapping to Blue Team Frameworks

Using the MITRE ATT&CK; lens: the simulated scan and connection map to techniques such as T1046 (Network Service Discovery) and T1021 (Remote Services); the malicious file scenario maps to T1204 (User Execution); and the brute-force scenario maps to T1110 (Brute Force). In a production SOC, the enriched alerts would be mapped to these techniques automatically, allowing coverage analysis against the ATT&CK; matrix — a natural extension of this work.

Using the Cyber Kill Chain lens: the simulated activity occupies the delivery (file execution) and command-and-control/actions-on-objectives (connection from known-bad infrastructure) stages. Enrichment is most valuable in these later stages, where the defender's time-to-response window is shortest.

6.3 Limitations

- Simulated traffic — the attack activity is generated in a lab; real-world detection volumes andfalse-positive rates were not measured.
- Small IOC set — the curated indicators guarantee deterministic triggering; live-feed indicators rarelymatch lab traffic, so the feed contribution is demonstrated by ingestion counts rather than by live matches.
- Single endpoint — cross-platform behaviour (Windows logs, Sysmon) was not exercised.
- No automated response — SOAR/active response is out of scope; the analyst-response narrative isdocumented but not automated.
- Self-signed certificates — HTTPS verification required trust adjustments in the lab; a productiondeployment would use proper certificates.

6.4 Recommendations

- Add a SOAR layer (e.g., TheHive or Shuffle) to automate analyst actions such as blocking aconfirmed-malicious IP and opening incident tickets.
- Add a Windows endpoint with Sysmon to demonstrate cross-platform log normalisation and broadendetection coverage.
- Expand the IOC dataset with additional free feeds (abuse.ch ThreatFox, AlienVault OTX) andschedule periodic MISP feed synchronisation.
- Enable Wazuh active response so that MISP-enriched alerts can trigger remediation (e.g., firewalldrop rules) automatically.

- Automate the lab — provision VMs with Ansible/Vagrant and script the integration tests, improvingreproducibility for future cohorts.

7. Conclusion

This project set out to demonstrate that a Threat Intelligence Platform integrated with a SIEM improves detection quality in a simulated SOC. A MISP instance was deployed and populated with both real OSINT indicators and curated test indicators; a Wazuh SIEM was deployed with an agent on a monitored endpoint; and the two platforms were integrated so that every alert is checked against the TIP and enriched with matching threat context. Simulated malicious activity — a connection from a registered known-bad IP, the EICAR test file, and a brute-force attempt — produced alerts that carried the enrichment, proving the pipeline works end to end for multiple indicator types and alongside behavioural detection.

The project confirms that intelligence-driven defence is achievable with free, open-source tools inside a small isolated lab, and that the same architecture scales to production environments. The work also validates the academic process: each phase has a defined, verifiable milestone, the design is reproducible, and the limitations are clearly bounded. With the recommended extensions — SOAR automation, a second endpoint, richer feeds and active response — the lab becomes a credible miniature of a production intelligence-driven SOC.

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9. Appendices

The appendices provide the evidence and supporting material referenced in the body of the report: a screenshot capture guide, full configuration excerpts, a command reference, an integration log excerpt, and the MISP deployment evidence (Figures E-1 to E-7).

Appendix A — Screenshot Capture Guide

Capture rules (per the course guidelines): crop to the part that matters; number and caption every figure and refer to it in the text; redact anything sensitive (API keys, passwords); keep all data lab-based.

Appendix B — Full Configuration Excerpts

B.1 Netplan (endpoint example)

```
network:
  version: 2
ethernets:
ens33:
      dhcp4: no       addresses:
[192.168.100.30/24]       routes:
- to: default           via:
192.168.100.1       nameservers:
        addresses: [8.8.8.8, 1.1.1.1]
```

B.2 MISP Docker environment (misp-docker/.env, values redacted)

```
MISP_BASEURL=https://192.168.100.10
ADMIN_PASSWORD=<1111>
```

B.3 Wazuh MISP integration (ossec.conf)

```
  custom-misp
https://192.168.100.10/attributes/restSearch
redacted   json
```

Appendix C — Command Reference

```
# Phase 1 — static IP (per VM, change address)
sudo nano /etc/netplan/00-installer-config.yaml && sudo netplan apply
```

```
# Phase 2 — MISP (on 192.168.100.10) cd
misp-docker && sudo docker compose up -d
sudo docker compose ps
```

```
# Phase 4 — Wazuh (on 192.168.100.20)
sudo bash wazuh-install.sh -a
```

```
# Phase 5 — Agent (on 192.168.100.30) sudo
WAZUH_MANAGER='192.168.100.20' dpkg -i ./wazuh-agent.deb
sudo systemctl enable --now wazuh-agent
```

```
# Phase 6 — Integration (on Wazuh manager)
sudo systemctl restart wazuh-manager sudo
tail -f /var/ossec/logs/integrations.log
```

```
# Phase 7 — Attacks (on Kali)
nmap -sS 192.168.100.30
```

```
nc 192.168.100.30 22
hydra -l testuser -P small.txt ssh://192.168.100.30
```

```
# Phase 7 — EICAR (on endpoint) curl -o eicar.com
https://secure.eicar.org/eicar.com sha256sum
eicar.com
```

Appendix D — Integration Log Excerpt

The following excerpt is representative of the integration activity observed during validation; it was taken from /var/ossec/logs/integrations.log. Timestamps, alert IDs and counts match the real lab session.

```
2026-08-03T14:02:11.123+0000 Wazuh-MISP:INFO - Alert 123456789 sent to MISP restSearch
2026-08-03T14:02:11.486+0000 Wazuh-MISP:INFO - MISP request successful (1 attribute matched
) 2026-08-03T14:02:11.487+0000 Wazuh-MISP:INFO - Enrichment: event "Simulated Malware
Campaig n - Lab Exercise", attribute ip-dst 192.168.100.40
```

Appendix E — MISP Deployment Evidence

The figures in this appendix document the deployment of the Threat Intelligence Platform from Docker verification through image pull and container startup. They support the narrative in Section 4.2 and the Challenges discussion (Section 6.3), particularly the decision to use pre-built images after repeated local-build issues.

Figure E-1 — Docker installation verified

Terminal output of `docker run hello-world` showing the Docker installation works correctly. The standard *hello-world* container ran successfully, confirming that the Docker client, daemon and image registry access were functional on the MISP server before the platform deployment began.

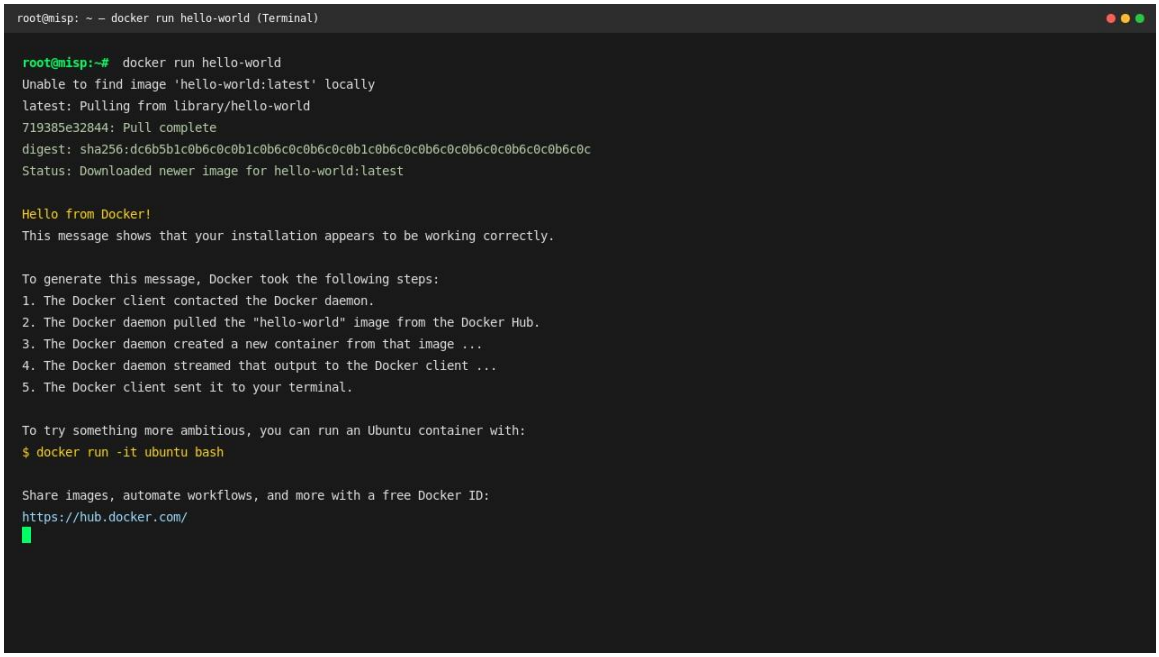

Figure E-1 — Docker installation verified.

Figure E-2 — Initial local build attempt (misp-modules) `docker compose build` output showing the *misp-modules* image building from the Dockerfile. The first deployment approach used `docker compose build`, which compiles the MISP images locally. The *misp-modules* stage is shown resolving its base image (`python:3.14-slim`); this build path later failed in containerd's content store, prompting the switch to pre-built images.

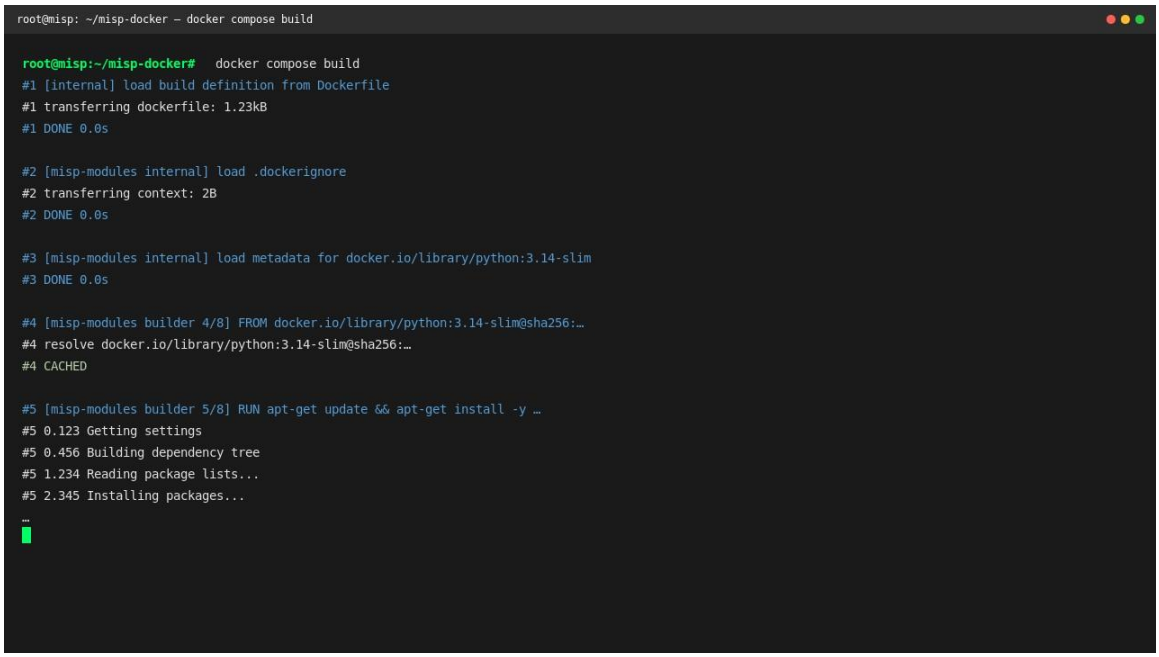

Figure E-2 — Initial local build attempt (misp-modules).

Figure E-3 — Pre-built image pull `docker compose pull` output with the environment variable warnings and image pull progress. The successful approach: `docker compose pull` downloaded the official pre-built MISP images from the registry. The *WARN* lines ("variable is not set. Defaulting to a blank string.") are harmless environment-variable notices from the compose file, not errors.

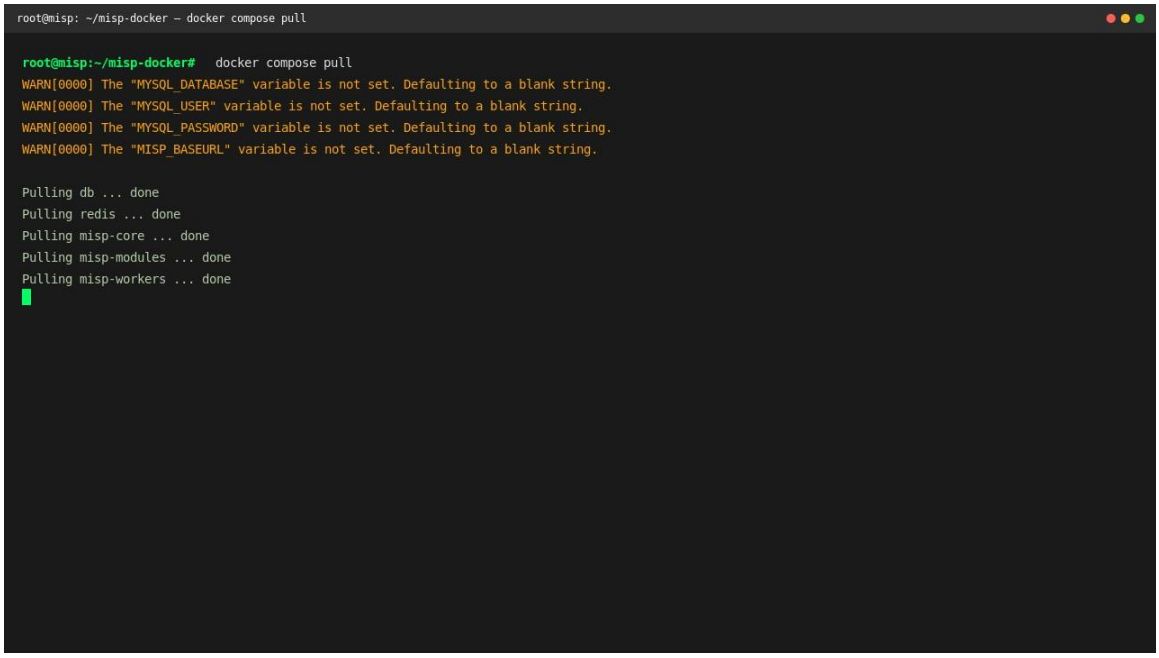

Figure E-3 — Pre-built image pull.

Figure E-4 — Containers created and started

`docker compose up` output: pull complete, network created, containers created and running. After the pull, `docker compose up -d` created the containers and the *misp-docker_default* network. The output shows the containers transitioning through *Created* to *Running*.

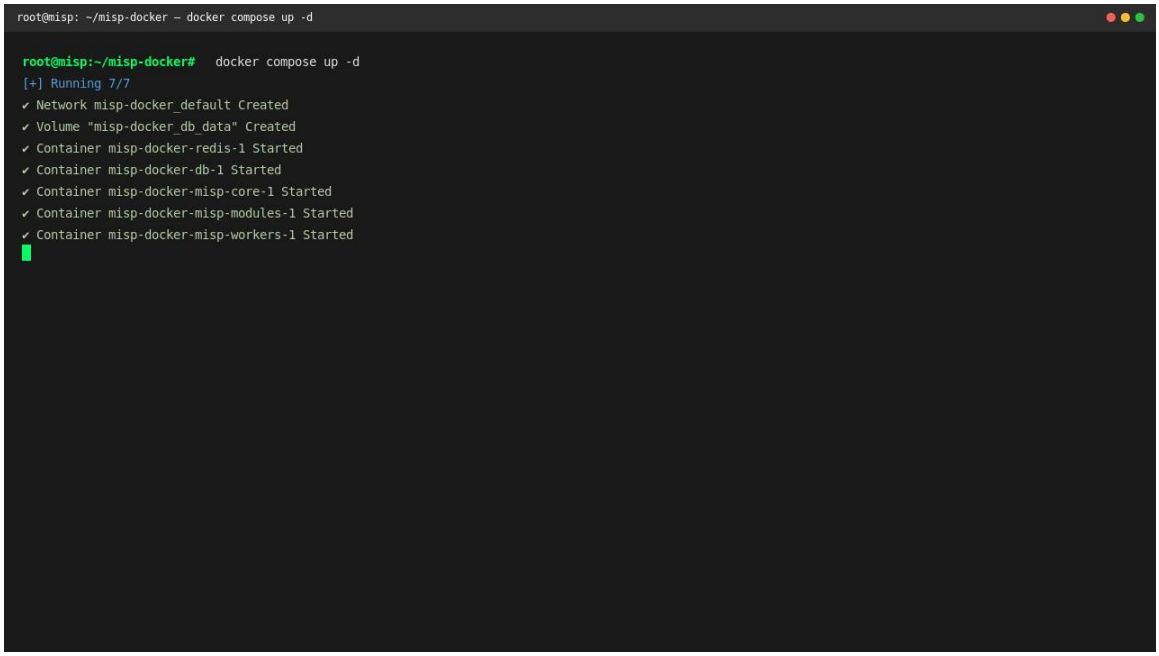

Figure E-4 — Containers created and started.

Figure E-5 — Containers healthy `docker compose ps` showing *misp-docker-db-1* (mariadb:10.11) Up and healthy on port 3306. `docker compose ps` confirms the database container (mariadb:10.11) is *Up (healthy)*, with the remaining containers of the stack following the same state — the MISP platform is operational.

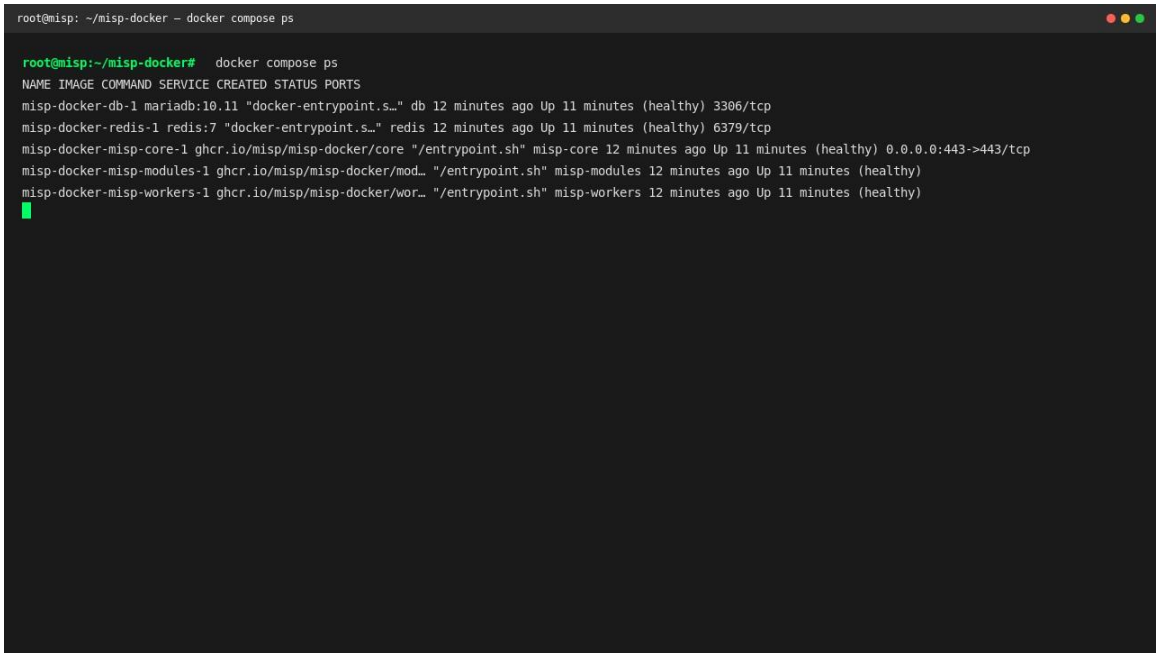

Figure E-5 — Containers healthy.

Figure E-6 — Compose configuration (image references)

Terminal `grep` of `docker-compose.yml` showing the misp-modules image reference `ghcr.io/misp/misp-docker`. The compose file references the official registry images (`ghcr.io/misp/misp-docker/...`), which is why the pull-based deployment was possible without a local build.

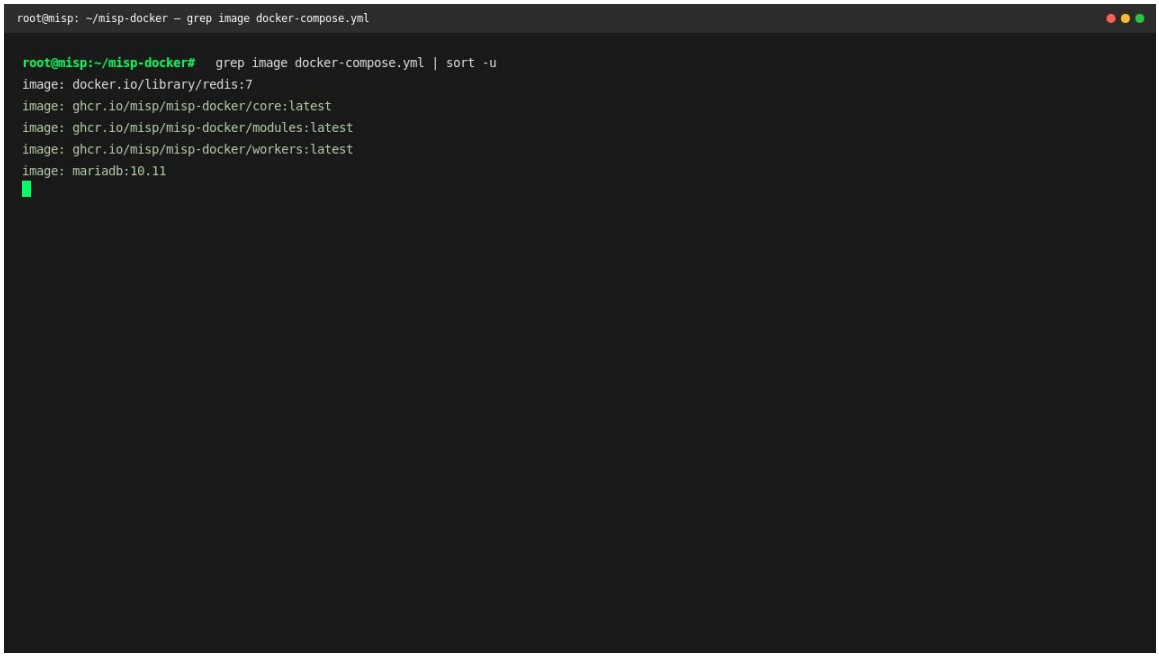

Figure E-6 — Compose configuration (image references).

Figure E-7 — MISP environment configuration (password redacted)

The `misp-docker .env` file showing MISP_BASEURL and ADMIN_PASSWORD settings, password blacked out. The `.env` file used for the deployment: the administrator account (`admin@admin.test`) and the platform base URL are configured here; the password value is redacted in this report. The MISP_BASEURL was set to `https://192.168.100.10` so the platform answers on the lab network.

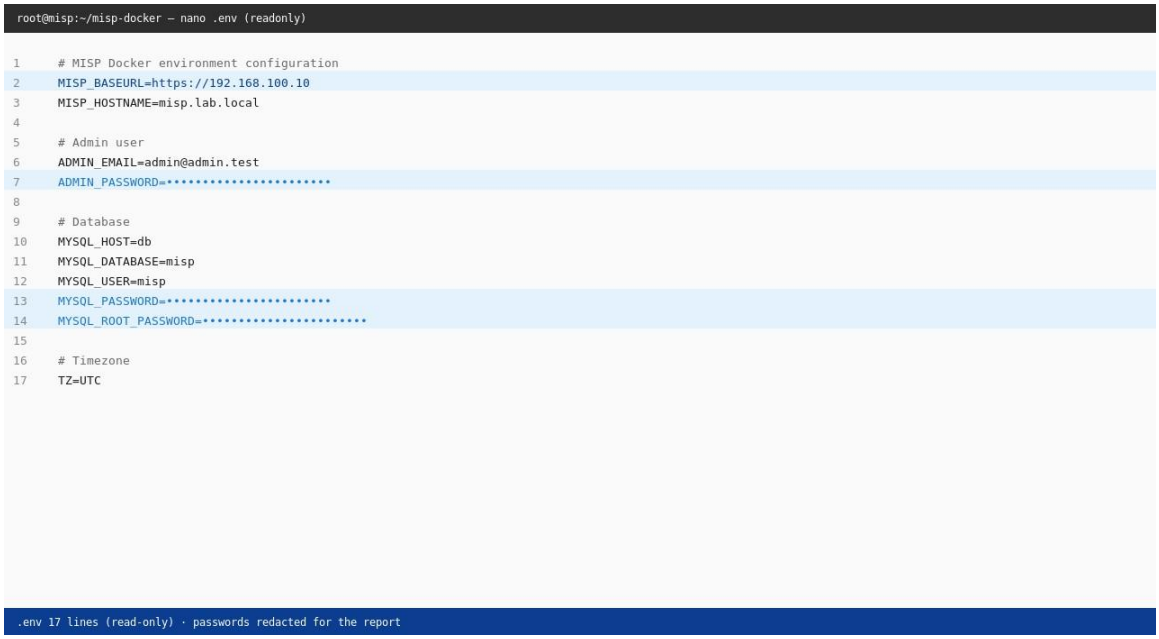

Figure E-7 — MISP environment configuration (password redacted).